LAUNCH THE PROJECT - 03

Flame Alarm System

(Intermediate level Project)

On AYNOP® UNO Launchpad Kit



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1. Overview

This project demonstrates how to build a **Flame Alarm System** using the Arduino® UNO R4 Minima.

The system detects flame using a flame sensor module. Once a flame is detected, the alarm is **latched ON** and remains active until the reset push button is pressed.

The system provides:

- Audible alert using buzzer
- Visual alert using onboard LED
- Status display on a 16×2 I2C LCD

You will learn to:

- Interface a flame sensor module.
- Use digital input detection.
- Implement latch-based alarm logic.
- Use a reset push button with external pull-up.
- Display system status on an I2C LCD

This project demonstrates a basic **fire detection alarm system** used in safety monitoring applications.

2. Components Required

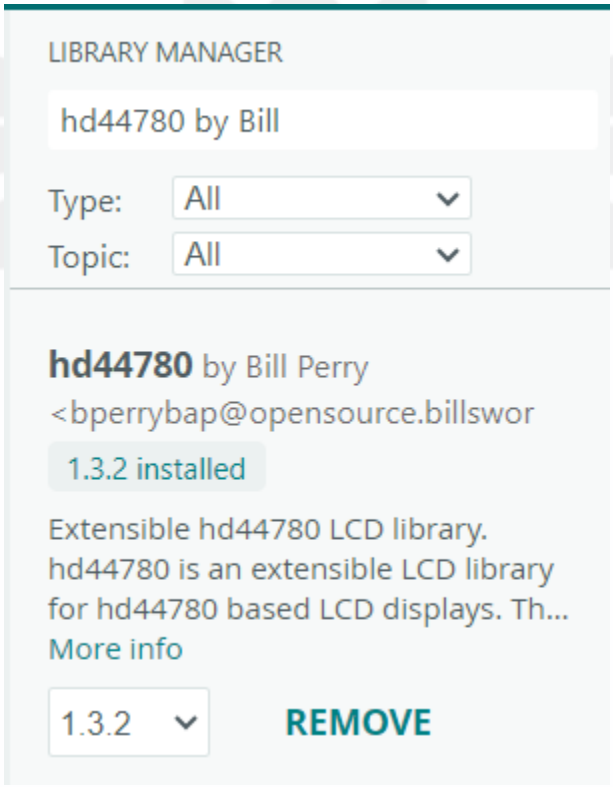
- Arduino® UNO R4 Minima board
- USB Type-C data cable
- 1 × 16x2 LCD with I2C adapter
- Flame sensor module
- Active buzzer
- Push button (reset switch)
- Jumper wires
- Breadboard

3. Software Required

- Arduino IDE (v2.3.7 or later)
- Required library – 02: **hd44780** for 16x2 I2C LCD Display

Installing the Library:

- If the **hd44780** library is already installed on your system, you may skip this step and proceed to Section 4.
- Else, Open Arduino IDE.
- Go to **Sketch → Include Library → Manage Libraries**
- Search for **"hd44780"** (by Bill Perry).
- Install the library.



Note:
We assume the **Arduino UNO R4 Minima board package** is already installed on your machine, as explained in the 00_Getting_Started/00_GettingStarted_Arduino_R4_Minima guide. If it is not installed, please refer to that document and complete the installation before proceeding.

4. Hardware Setup

This section explains how to connect the flame sensor, LCD, buzzer, and reset button.

4.1 Wiring Diagram

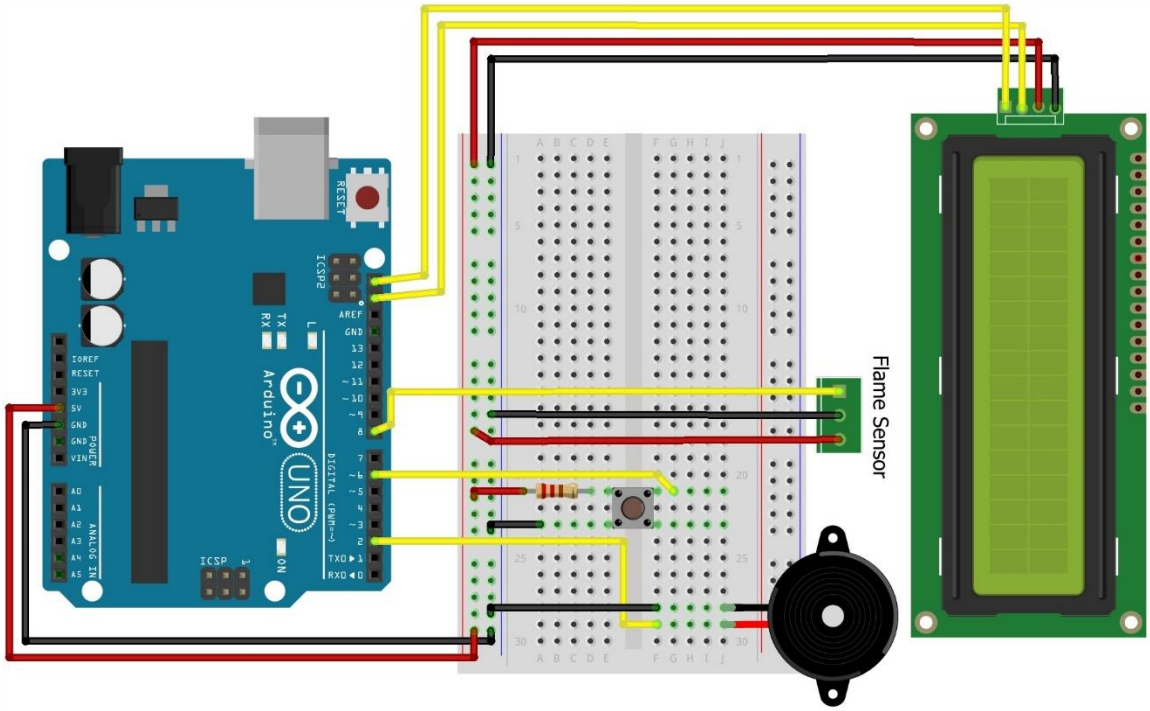


Figure 4.1 – Wiring diagram for flame alarm system

Tip: I2C uses the same two wires (SDA, SCL) for multiple devices. Each device has a unique **I2C address**.

Tip: Always disconnect the USB cable before making or changing hardware connections.

5. Principle – How It Works

The flame sensor detects infrared light emitted by a flame. Most flame sensor modules give a LOW signal when flame is detected and HIGH when no flame is present.

The Arduino continuously reads this signal from the sensor.

Alarm Operation

When a flame is detected:

- Arduino activates the buzzer and onboard LED
- LCD displays “FLAME DETECTED – RESET REQUIRED”
- Alarm remains ON even if the flame disappears

This is called a **latched alarm**, ensuring the user does not miss a fire event.

Reset Operation

A reset push button is provided to clear the alarm.

When the reset button is pressed:

- Alarm turns OFF
- LED and buzzer stop
- LCD returns to “No Flame” status

The system then continues monitoring for flame again.

This project demonstrates a simple **fire detection and safety alert system** used in real-world monitoring applications.

6. Procedure – Steps to Run

1. **Build the Circuit**
   a. Connect the flame sensor, LCD, Push Button and buzzer as shown in the wiring diagram (Figure 4.1).

2. **Connect the Board**
   a. Use a USB Type-C data cable to connect your UNO R4 Minima to your computer.

3. **Open the Project Code**
   a. Simply **double-click** the file *03_Flame_Alarm_System.ino* in the project folder, and it will open directly in the Arduino IDE (if installed).

4. **Confirm Board Selection**
   a. The IDE usually auto-detects the UNO R4 Minima if the package is installed.
   b. If not installed, refer to the *00_Getting_Started/00_GettingStarted_Arduino_R4_Minima* document to install the necessary board package.
   c. Verify that *Arduino UNO R4 Minima* is selected in the IDE's board selector (top toolbar).

5. **Upload the Code**
   a. Click the **Upload** button (arrow icon) in the top-left corner of the IDE.
   b. Wait until the console displays "**Done uploading**."

6. **Observe the Behaviour**
   a. The LCD will display the current system status.
   b. When no flame is present, the LCD shows "Status: No Flame".
   c. When a flame is detected, the LCD displays "FLAME DETECTED" and "RESET REQUIRED".
   d. The buzzer and onboard LED turn ON immediately when flame is detected.
   e. The alarm remains ON even if the flame is removed.
   f. Press the reset push button to turn OFF the alarm and return to normal monitoring mode.

Tip: Use a lighter or candle flame at a safe distance and observe detection.

7. Expected Output

- **Normal Condition (No Flame):**
  - LCD displays **Status: No Flame**.
  - LED OFF.
  - Buzzer OFF.

- **Flame Detected:**
  - LCD displays **FLAME DETECTED** and **RESET REQUIRED**.
  - LED turns ON.
  - Buzzer turns ON.
  - Alarm remains ON (latched).

- **Reset Condition:**
  - Press reset push button.
  - LCD returns to **No Flame**.
  - LED OFF.
  - Buzzer OFF.

8. Code

The source code for this project is included in the downloaded folder:

📂 uno-launchpad-kit/02_Intermediate_Projects/03_Flame_Alarm_System/03_Flame_Alarm_System.ino

👉 **Tip:**

- To open the project, simply **double-click the .ino file**. If the Arduino IDE is installed, it will launch automatically and load the code.
- If you **haven't installed the Arduino IDE yet**, please refer to:
  📂 uno-launchpad-kit/00_Getting_Started/00_GettingStarted_Arduino_R4_Minima
  to **download and install it.**

8.1 Function References

- setup() – Runs once when the board is powered on or reset. Used to initialize sensors, LCD, and I/O pins.
- loop() – Runs continuously after setup () finishes. Reads sensor values and updates the LCD at fixed intervals.
- lcd.begin(cols, rows) – Initializes the 16×2 I2C LCD.
- pinMode(pin, mode) – Configures a pin as INPUT or OUTPUT.
- digitalWrite(pin, value) – Turns the LED and buzzer ON or OFF
- lcd.backlight() – Turns ON the LCD backlight for better visibility.
- lcd.clear() – Clears all text currently displayed on the LCD.
- lcd.setCursor(column, row) – Sets the cursor position on the LCD screen.
- lcd.print() – Displays text or numeric values on the LCD.
- digitalRead(pin)– Reads signal from flame sensor and reset button.
- delay(ms) – Pauses program execution for a fixed time interval between sensor updates.

📚 **Library Reference:**

📎 Wire Library — I2C communication library used internally by hd44780.
📎 hd44780 — I2C LCD control library used to interface the 16×2 display.

📚 **For more details and advanced usage, visit:**
📎 Arduino Language Reference — The official guide for all Arduino functions.

9. Troubleshooting Tips

- **Error while compiling (hd44780 library not found)?**
  - Open Arduino IDE → Library Manager.
  - Search for "hd44780 (by Bill Perry)" and install it.
  - Restart the IDE and compile again

- **LCD backlight ON but no text?**
  - Adjust the small **blue potentiometer** on the I2C adapter to set contrast.

- **LCD backlight not ON?**
  - I2C adapter boards have a small **jumper near the backlight pins**.
  - Ensure this jumper is connected; otherwise, the backlight will stay OFF

- **LCD shows nothing?**
  - Check wiring: VCC → 5V, GND → GND, SDA → D18, SCL → D19.
  - Verify the **I2C address** (0x27 or 0x3F)

- **Upload error in Arduino IDE?**
  - Verify that the correct board (**Arduino UNO R4 Minima**) is selected in the IDE.
  - Check that the correct **COM port** is chosen.

- **Buzzer not sounding?**
  - Check buzzer polarity and connection to D6.

- **Alarm always ON or Flame not detected?**
  - Flame sensor may be too sensitive.
  - Rotate onboard potentiometer to adjust sensitivity.

- **Reset button not working?**
  - Check button wiring to D4 and GND.
  - Ensure button uses external pull up resistor.
  - Keep wires short and connections tight to reduce noise

- **Still not working?**
  - Recheck all wiring connections carefully.
  - Disconnect power and rebuild the circuit step-by-step.
  - Test each module (LCD, Buzzer, flame Sensor) individually with help of basic projects.

10. License

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11. Support & Feedback

We value your feedback and are happy to assist with any questions, troubleshooting, or suggestions you may have.

Email: *support@aynop.com*

When sending an email, please include your kit name (AYNOP® UNO Launchpad Kit) and, if applicable, the project name in the subject line. This will help our team respond faster and more accurately. We aim to respond to all queries within 2–3 business days. Your feedback helps us improve our products and create even better learning experiences for all Arduino beginners.